

KARTHIK KOTA

Software Engineer || AI/ML Engineer

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PROFESSIONAL SUMMARY

Software Engineer with 3+ years of experience building production-grade AI/ML systems, scalable backend services, and cloud data platforms across enterprise CRM and data-intensive domains. Architected **RAG pipelines**, **LangChain**, and **OpenAI API** integrations on AWS SageMaker processing millions of transactions at sub-100ms latency. ↑ Designed event-driven microservices, FastAPI REST APIs, and Apache Spark and Kafka ETL pipelines ingesting 200GB+ daily at 99.9% uptime. ↑ Deployed **XGBoost**, CNN, and LSTM models end-to-end using **MLflow**, Docker, and GitHub Actions CI/CD across AWS and GCP infrastructure. ↓ Zero SOC 2 compliance violations across all production ML pipelines.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, Bash

AI & GenAI: **RAG**, **LangChain**, **OpenAI API**, Hugging Face Transformers, AWS Bedrock, Claude API, NER, Prompt Engineering, Transformer Architecture

Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost, LightGBM, Keras, CNN, LSTM, Random Forest, Feature Engineering, Hyperparameter Tuning

Backend & System Design: Data Structures & Algorithms, System Design, REST APIs, FastAPI, Microservices, OOP, Unit Testing, Agile/Scrum, Git, Code Review

MLOps & DevOps: **MLflow**, Docker, GitHub Actions, FAISS, Pinecone, CI/CD, Model Versioning, Experiment Tracking

Cloud & Data Engineering: AWS (EC2, S3, Lambda, SageMaker, Bedrock, IAM, CloudWatch), GCP (Vertex AI, BigQuery), Terraform, Apache Spark, Kafka, Spark Streaming, ETL Pipelines, Pandas, NumPy, Tableau, Power BI

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, NoSQL

Security & Compliance: SOC 2, HIPAA, IAM, Data Encryption, Policy Enforcement

PROFESSIONAL EXPERIENCE

Software Engineer - AI & ML Systems Salesforce

Dallas, TX
Jun 2025 – Present

RAG · LangChain · OpenAI API · AWS SageMaker · MLflow · XGBoost · Apache Spark · Kafka · Docker · GitHub Actions · FastAPI · PostgreSQL · DynamoDB · SOC 2 · IAM · Tableau · Power BI

- Architected a production LLM-powered CRM intelligence platform using RAG, LangChain, and OpenAI API on AWS SageMaker — automated data extraction across 500K+ customer records, slashing manual processing time by 45% and reducing operational overhead by 60%.
- Engineered a high-throughput XGBoost predictive scoring engine processing 1M+ CRM events monthly at sub-100ms latency — accelerated ML iteration cycles by 50% using MLflow Experiment Tracking and Model Versioning.
- Re-architected legacy monolithic into distributed, event-driven microservices using Kafka and Docker on AWS — tripled system throughput, reduced p99 latency from 800ms to 150ms, and improved scalability and platform reliability by 40%.
- Drove database design and optimized PostgreSQL and DynamoDB schemas supporting 10M+ CRM records at sub-50ms query times — reduced database load by 60% through intelligent caching and query indexing strategies.
- Built GitHub Actions CI/CD pipelines and Docker-containerized Apache Spark ETL workflows ingesting 200GB+ daily at 99.9% uptime — compressed deployment cycles from 3 days to 4 hours.
- Delivered 3 major product releases end-to-end in Agile sprints with 95%+ unit test coverage, zero critical production bugs, and full SOC 2 and IAM compliance across all ML pipelines.

Software Engineer – AI & Data Platforms Publicis Sapient

Hyderabad, India
May 2021 – June 2023

Python · TensorFlow · PyTorch · scikit-learn · XGBoost · Apache Spark · Kafka · FastAPI · REST APIs · Docker · GitHub Actions · PostgreSQL · DynamoDB · LangChain · OpenAI API · MLflow · Terraform · AWS · Power BI · SOC 2 · IAM

- Promoted from Intern to Software Engineer within 12 months — owned AI-driven scientific data platforms processing instrument telemetry across global research facilities in 3 countries.
- Deployed CNN and LSTM models in TensorFlow and PyTorch for real-time sensor anomaly detection — cut equipment downtime alerts by 30%, improved signal accuracy by 22%, and saved 15+ inspection hours weekly.
- Engineered Apache Spark and Kafka pipelines ingesting 100K+ IoT events per minute; optimized PostgreSQL and DynamoDB queries by 50% on 5M+ record datasets — delivering sub-second data access globally.

- Led API design and built FastAPI backend services integrating ML inference into LIMS and ERP systems via REST APIs; automated CI/CD with GitHub Actions and Docker — reducing deployment cycles by 35% and production incidents by 40%.
- Architected a LangChain and OpenAI API RAG knowledge assistant over 500K+ scientific documents — accelerated compliance data retrieval by 60% and cut research turnaround time by 25% across global teams.
- Implemented MLflow Model Registry across 10+ ML models and enforced SOC 2 compliance via IAM, Data Encryption, and Terraform on AWS — achieving zero compliance violations across all production deployments.

ENGINEERING PROJECTS

AI-Powered Multi-Agent Autonomous System *RAG · LangChain Agents · OpenAI API · CrewAI · AWS Bedrock · Pinecone · FastAPI · Docker*

- Architected a multi-agent autonomous system using **CrewAI** and LangChain Agents orchestrating specialized AI agents for real-time research, analysis, and decision-making across enterprise workflows — reduced manual processing time by 65%.
- Integrated RAG over 1M+ enterprise documents with Pinecone vector search and OpenAI API achieving sub-200ms response times for 1K+ concurrent users on AWS Bedrock via FastAPI and Docker.
- Implemented agent memory, tool-use, and reflection loops improving task completion accuracy by 40%; enforced SOC 2 compliance via IAM and encrypted vector namespaces.

Enterprise Financial Fraud Intelligence Platform *OpenAI API · RAG · LangChain · BERT · FAISS · Pinecone · AWS SageMaker · MLflow · Docker · SOC 2*

- Architected a RAG fraud detection system integrating BERT embeddings with FAISS and Pinecone — processed 1M+ daily transactions at 95ms latency and 99.95% uptime on AWS EC2 via FastAPI and Docker.
- Reduced false positive fraud alerts by 24% using vector similarity search and XGBoost classifiers; MLflow Model Registry and GitHub Actions CI/CD enabled weekly zero-downtime releases.
- Cut MTTD by 52% via CloudWatch monitoring; enforced SOC 2 compliance through IAM, Data Encryption, and encrypted Pinecone namespaces.

Real-Time ML Feature Store & Model Serving Platform *MLflow · Apache Kafka · FastAPI · AWS SageMaker · Redis · Docker · GitHub Actions · PostgreSQL*

- Built an end-to-end real-time ML feature store using Apache Kafka and Redis serving 500K+ feature requests daily at sub-20ms latency — eliminating training-serving skew across 15+ production ML models.
- Designed FastAPI model serving layer on AWS SageMaker with A/B testing, shadow deployment, and canary rollouts; GitHub Actions CI/CD enabled zero-downtime weekly model releases.
- Implemented automated model drift detection and retraining triggers via MLflow Model Registry — maintaining 95%+ model accuracy across all production deployments with zero manual intervention.

EDUCATION

Master of Science — Computer and Information Science

May 2025

University of North Texas · Denton, TX

Bachelor of Technology — CS & Engineering (Artificial Intelligence)

May 2023

Vellore Institute of Technology · Vellore, India

CERTIFICATIONS

- **AWS Certified Machine Learning – Associate** — Amazon Web Services
- **TensorFlow Developer Certificate** — Google
- **Deep Learning Specialization** — Coursera – deeplearning.ai